SIGGRAPH ASIA 2025 HONG KONG 香港

Conference	**15 – 18 December 2025**
Exhibition	**16 – 18 December 2025**
Venue	**Hong Kong Convention and Exhibition Centre**

AniMaker: Multi-Agent Animated Storytelling with MCTS-Driven Clip Generation

Haoyuan Shi, Yunxin Li, Xinyu Chen,
Baotian Hu, Longyue Wang, Min Zhang
Harbin Institute of Technology, Shenzhen (HITSZ)

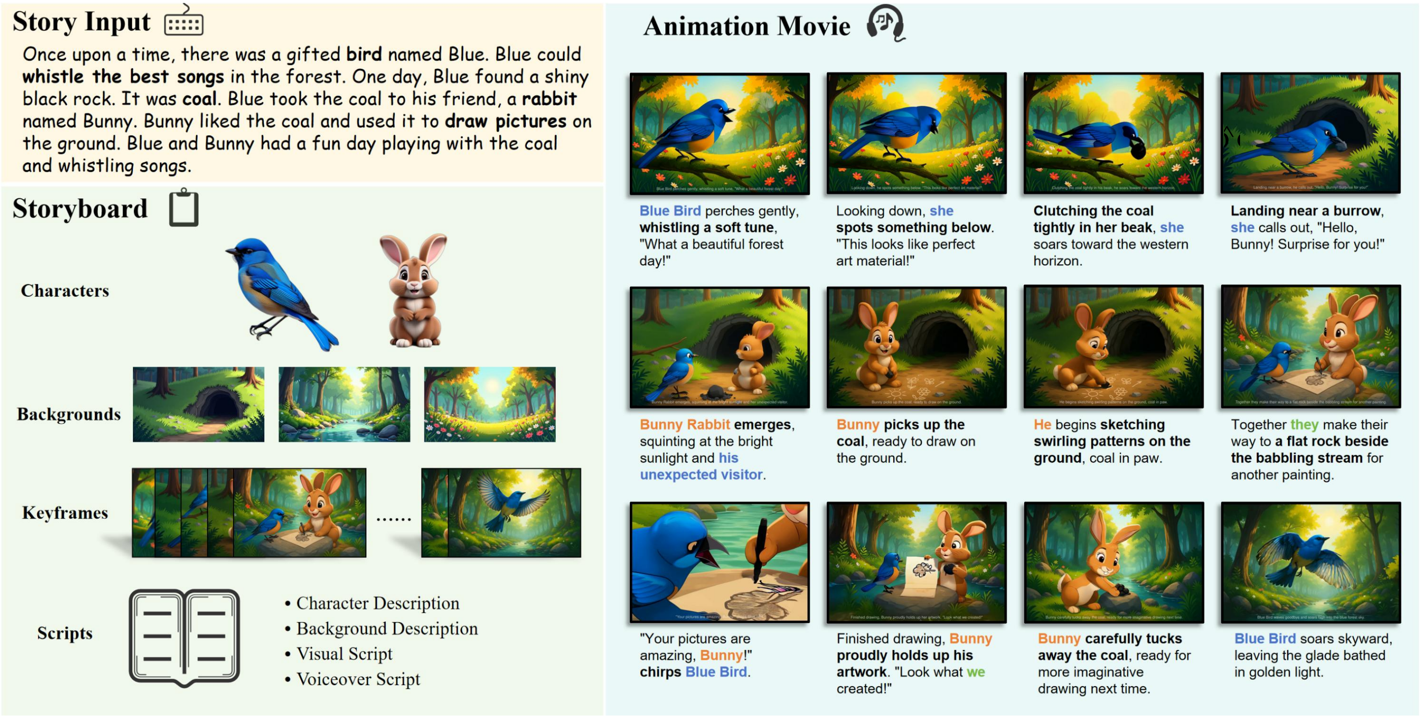
Sponsored by acm
Organized by koelnmesse

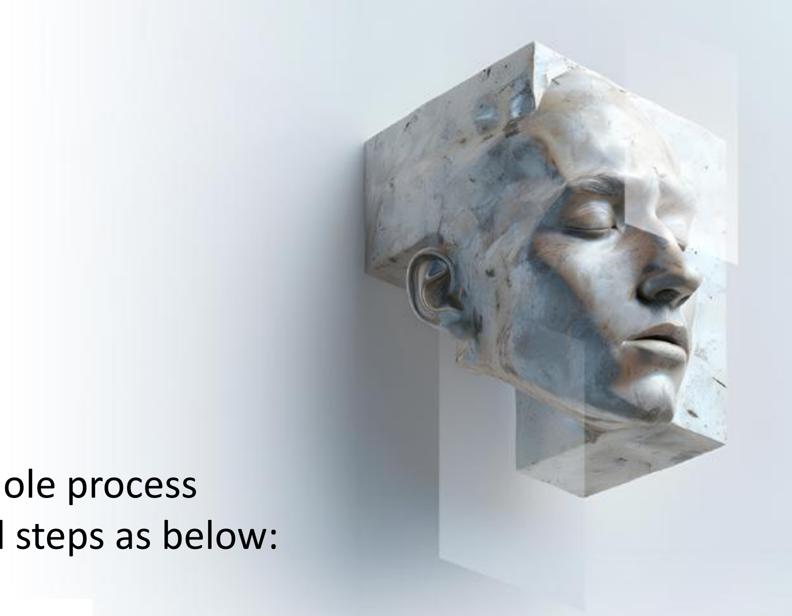
SIGGRAPH ASIA 2025 HONG KONG 香港

Animated Storytelling

Task Form

Given a short story or prompt,
the system automatically produces a complete animated video.





Typically, the whole process
may take several steps as below:

$$\mathcal{F} : \mathbf{T}_{\text{prompt}} \rightarrow \mathbf{V}_{\text{final}}$$

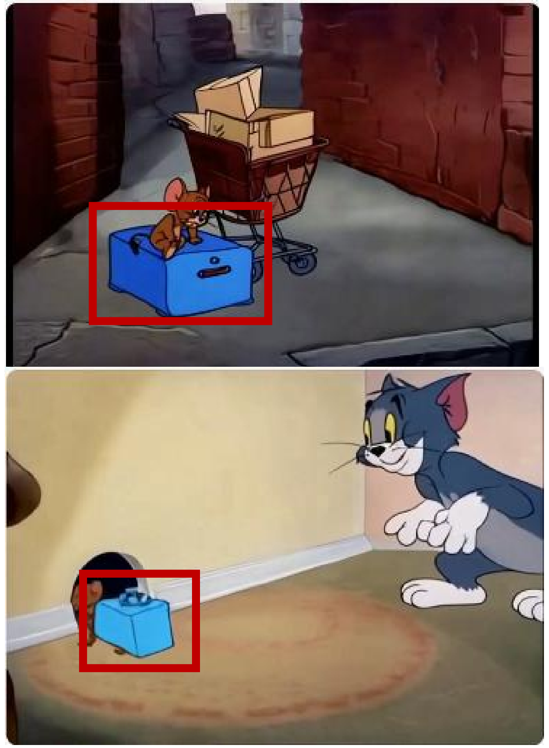
- **1. Script Planning ($\mathcal{P}_{\text{script}}$):**
  - Decomposes prompt into **ordered shots** ($\mathbf{shot}_k$).
  - Identifies **shot transitions** ($\mathbf{K}_{cut}$) requiring fresh visual guidance.
- **2. Storyboard Construction ($\mathcal{S}_{\text{board}}$):**
  - **Asset Banks:** Maintains consistent Character ($\mathcal{B}_{\text{char}}$) and Background ($\mathcal{B}_{\text{bg}}$) libraries.
  - **KeyFrame Generation:** Visualizes "anchor" frames ($\mathbf{F}_k^{\text{key}}$) at transition points.
- **3. Conditional Video Generation:**
  - **New Shot:** Generated from KeyFrame (G_K).
  - **Continuation:** Generated from the previous clip's last frame (G_C).
- **4. Final Production:** Sequence assembly and post-processing (audio/subs).

Animated Storytelling

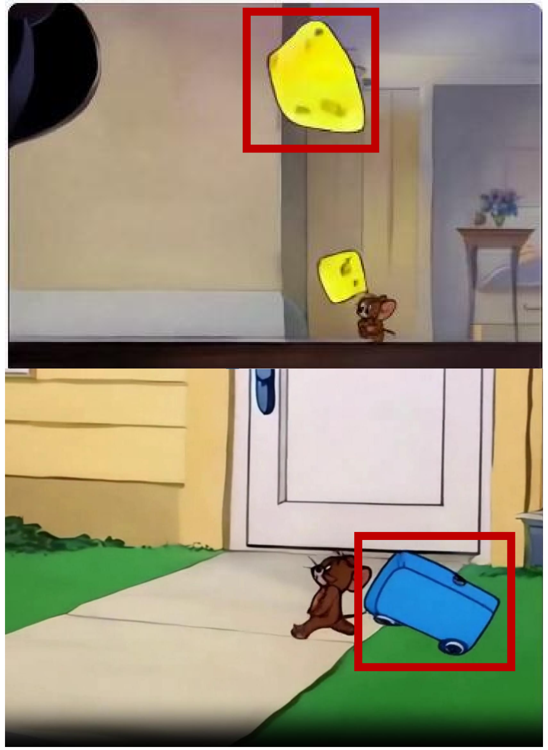
Challenges

Current methods struggle to maintain consistency in object appearance, physics, and backgrounds, especially in multi-shot video generation.

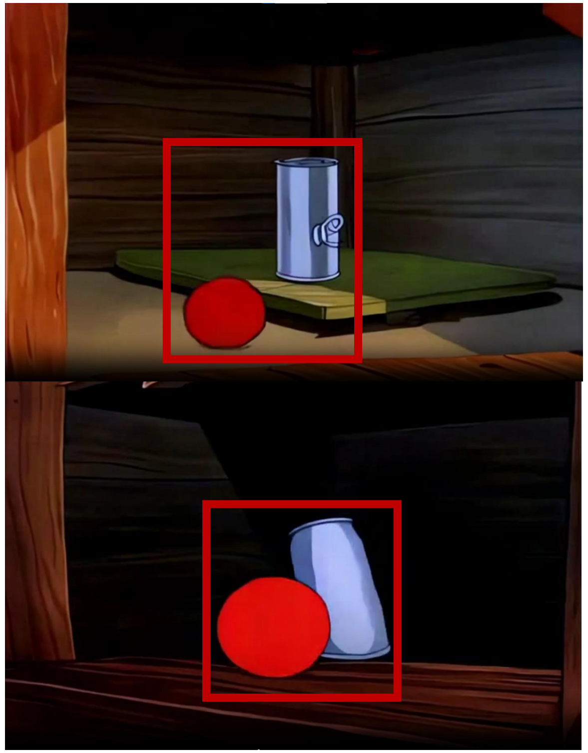
Inconsistent Object Appearance



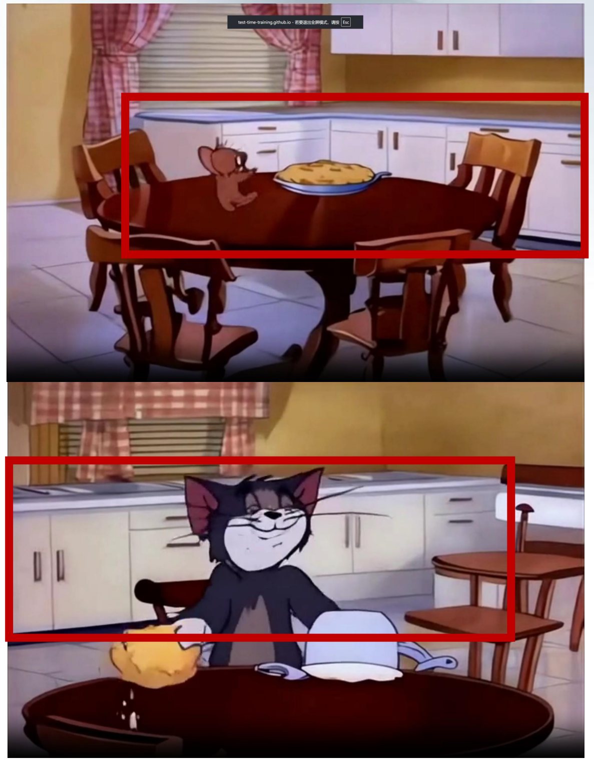
Poor Adherence to Physics



Inconsistent Scale Between Objects



Inconsistent Background Across Shots

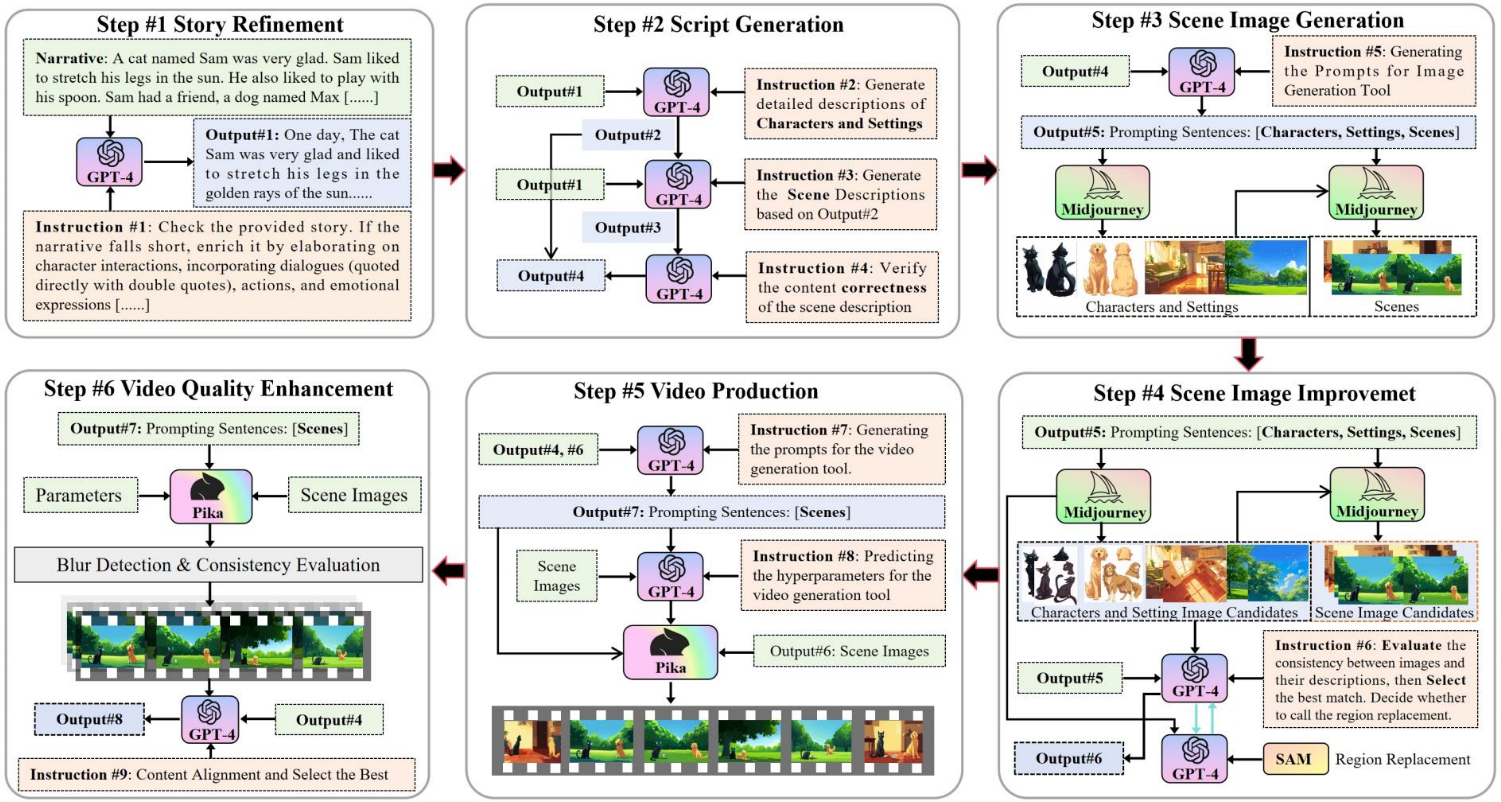


SIGGRAPH ASIA 2025 HONG KONG 香港

Animated Storytelling

Previous Methods

Generally follow a fixed pipeline: script → keyframes → video clips → final video composition



Rigid & Fragmented Structure

- Current pipelines map keyframes to fixed-length clips, ignoring the dynamic nature of storytelling.
- This leads to disjointed transitions and unnatural pacing, severing the expressive continuity required for complex actions.

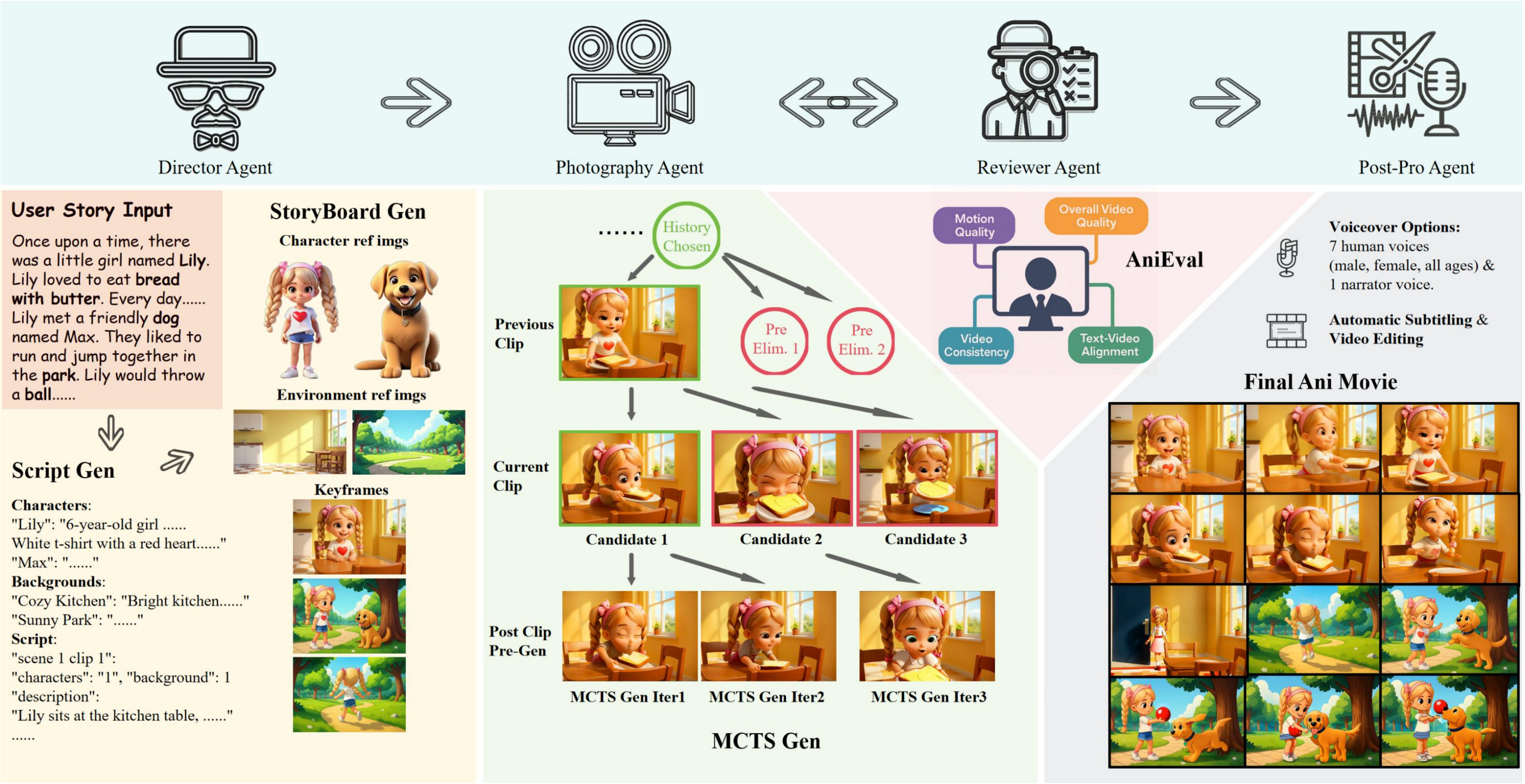
Vulnerability to Generation Instability

- Due to the inherent stochasticity of video generation models, the pipeline lacks robustness.
- A single flawed clip acts as a bottleneck, noticeably degrading the quality and coherence of the entire video.

SIGGRAPH ASIA 2025 HONG KONG 香港

AniMaker

MCTS-Driven Multi-Agent Framework Capable of Generating Globally Consistent Animations



- **Multi-Agent Collaboration**

Mirrors professional production workflows by coordinating four specialized agents—Director , Photography, Reviewer, and Post-Production.

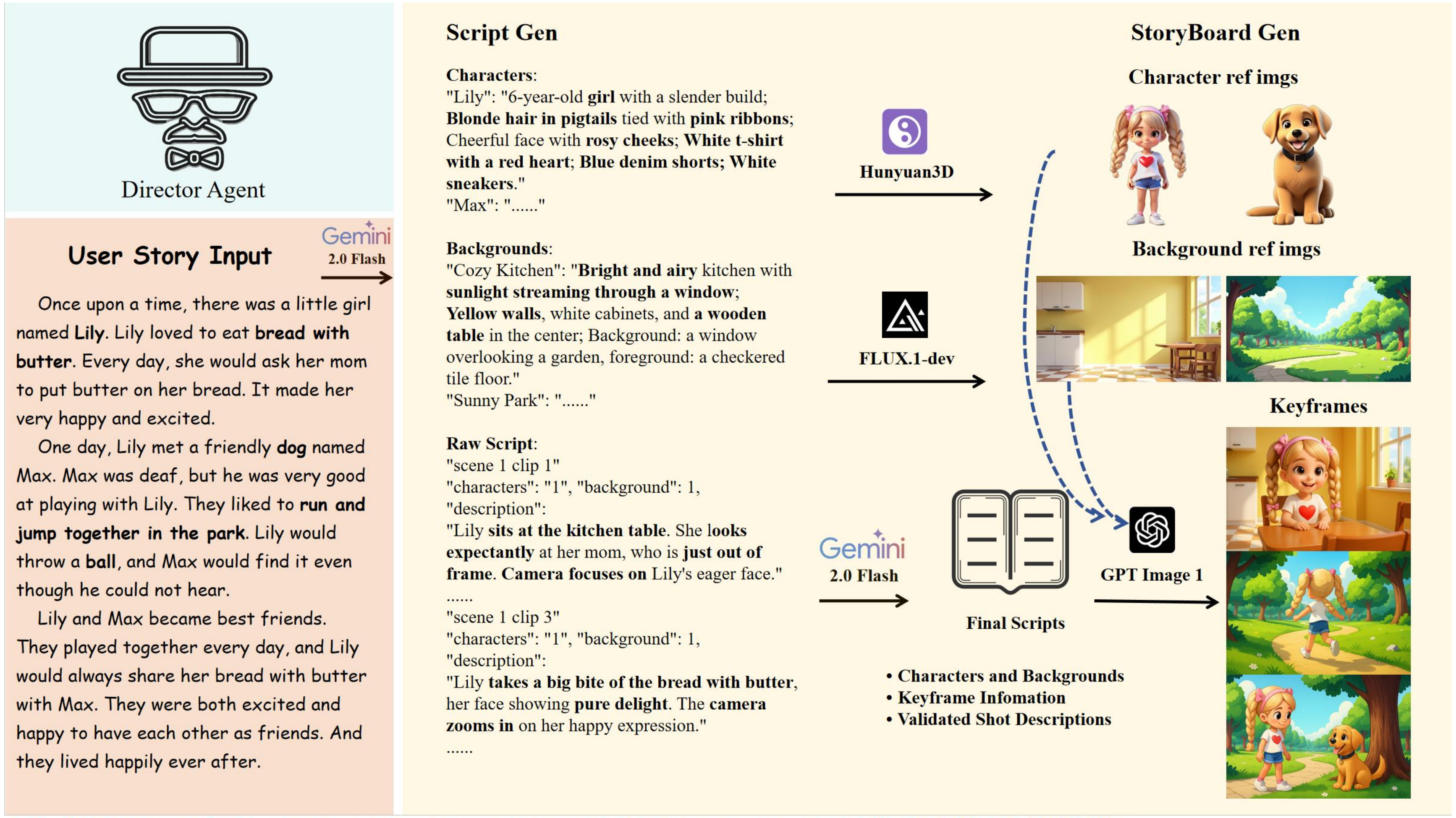
- **MCTS-Gen Strategy**

An efficient generation strategy inspired by Monte Carlo Tree Search that navigates the video candidate space, balancing exploration and efficiency.

- **AniEval Framework**

The first evaluation metric specifically designed for multi-shot animation, which assesses quality and consistency by analyzing each clip in the context of its neighbourhood shots.

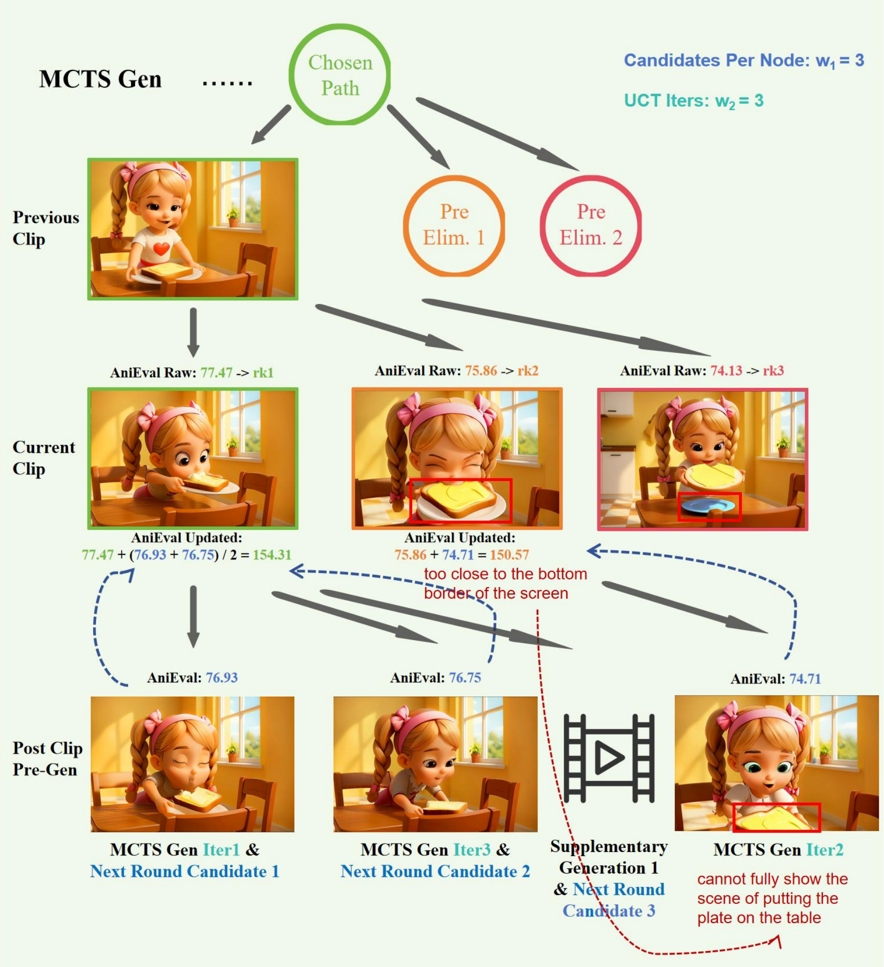
AniMaker Director Agent



Script Generation
Gemini 2.0 Flash analyzes the user story to generate structured character profiles, background descriptions, and a raw script with specific shot details.
Visual Bank Construction
The system builds a reference library by generating consistent 3D characters using Hunyuan3D and scenic backgrounds via FLUX.1-dev.
Storyboard Synthesis
GPT Image 1 combines the validated script and visual references to generate keyframes, ensuring visual consistency across the entire storyboard.

AniMaker

Photograph Agent & Reviewer Agent



MCTS-Gen Strategy

MCTS-Gen is a framework that adapts **Monte Carlo Tree Search** to optimize multi-clip video generation by modeling video sequences as tree paths and individual clips as nodes. Leveraging the Wan 2.1 generator and the AniEval scoring system, the method iteratively constructs coherent video sequences by systematically balancing the exploration of diverse generation spaces with the exploitation of high-quality narrative transitions.

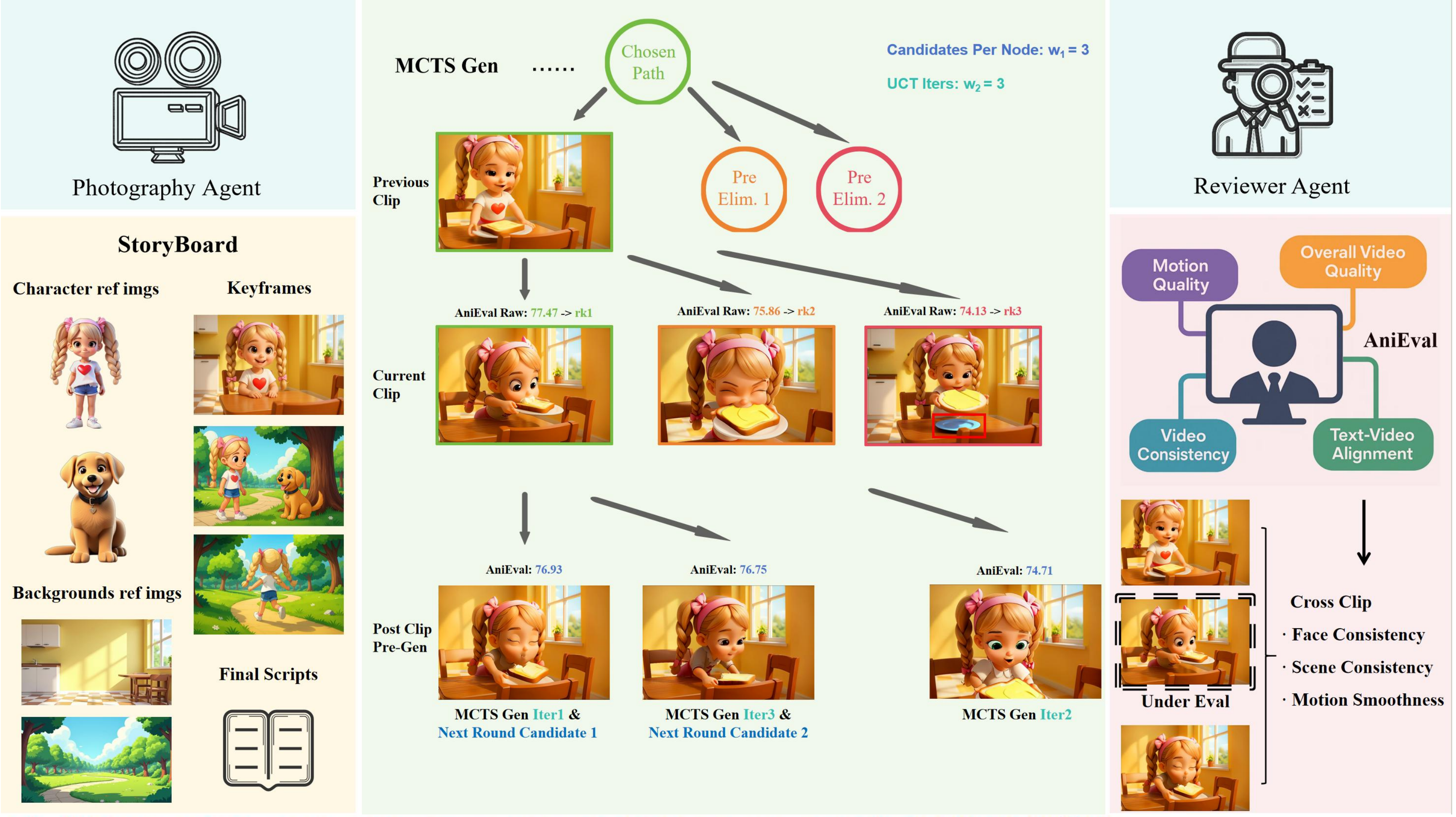
Expansion: The process begins by generating $\mathbf{w_1}$ initial candidate clips from the current terminal node of the chosen path, which are immediately scored and ranked via AniEval.
Simulation: The system performs $\mathbf{w_2}$ further expansions guided by a UCT score, which dynamically selects nodes for generation based on a trade-off between their current quality rank and the need for exploration (controlled by $\boldsymbol{\alpha}$).
Backpropagation: The evaluation scores of newly generated child clips are propagated upward, updating the parent nodes' values to reflect future potential.
Selection: Finally, the node with the highest AniEval score is permanently added to the chosen path, and the tree is prepared for the next iteration of the cycle.

$$UCT(node_j) = \frac{2.0}{rank(node_j)+1} + \alpha \cdot \sqrt{\frac{2.0}{child\_count(node_j)+1}}$$

SIGGRAPH ASIA 2025 HONG KONG 香港

AniMaker

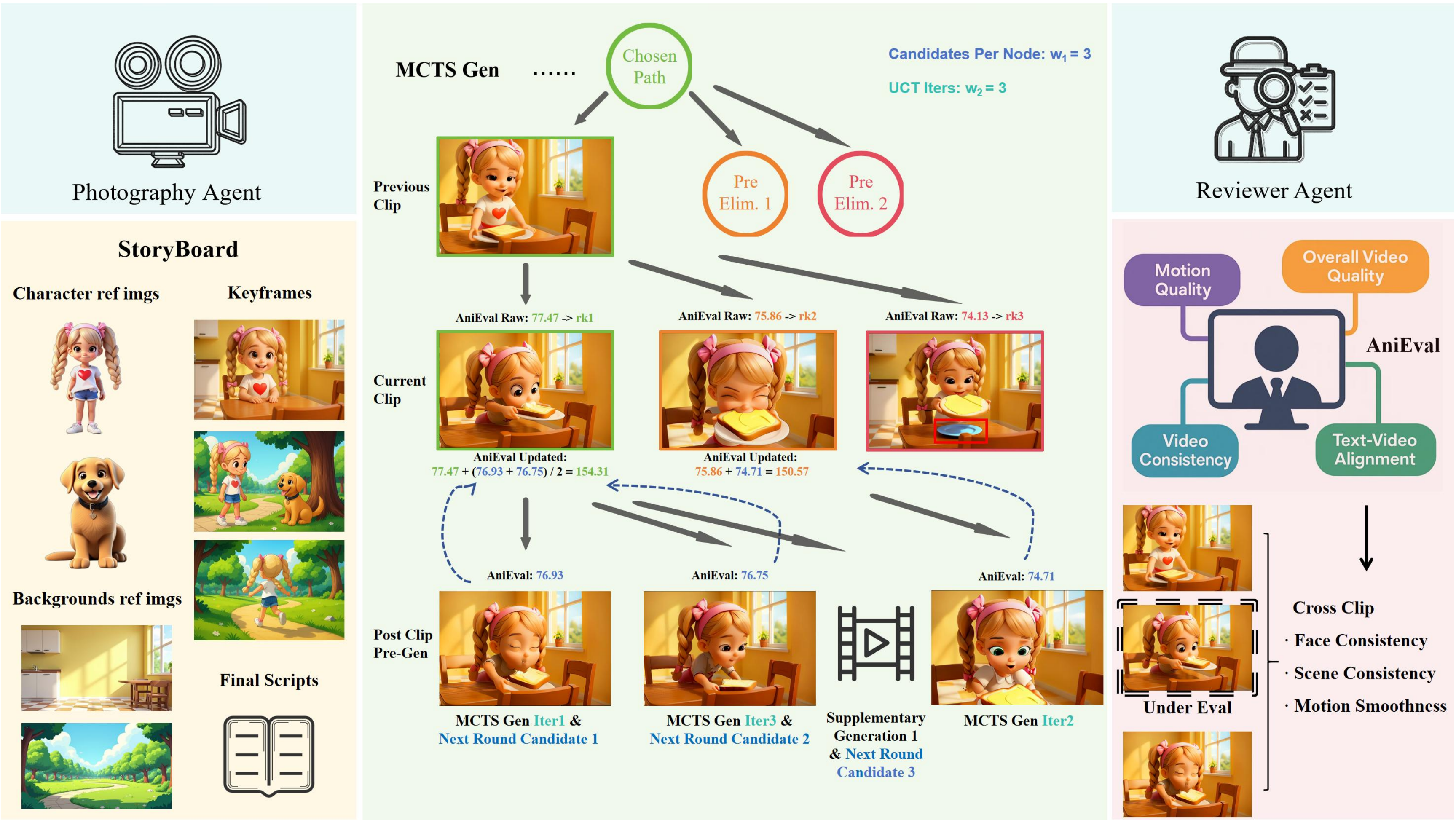
Photograph Agent & Reviewer Agent



- We define the search budget via hyperparameters $\mathbf{w_1}$ (expansion width) and $\mathbf{w_2}$ (simulation depth).
- Expansion: The Photograph Agent generates $\mathbf{w_1}$ candidate clips from the previous state. These are immediately filtered by the Reviewer Agent to prune obvious failures.
- Simulation: The system performs $\mathbf{w_2}$ look-ahead steps driven by the UCT algorithm. By balancing quality scores (exploitation) with visit rarity (exploration), UCT ensures the evaluation of diverse motion paths before final selection.

AniMaker

Photograph Agent & Reviewer Agent

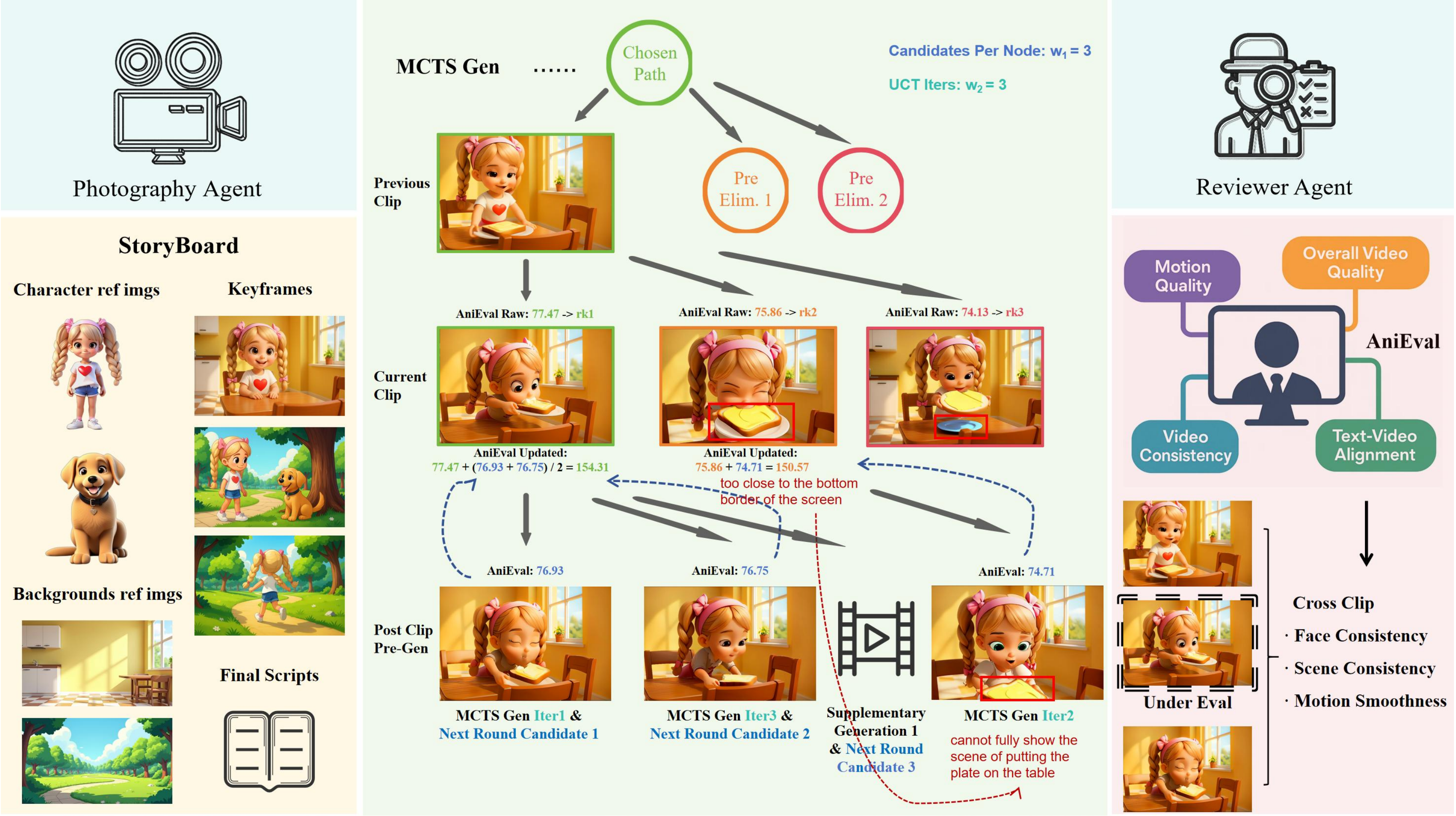


- As these new child nodes (seen at the bottom as MCTS Gen Iter1, Iter2, Iter3) are generated and scored, their values are backpropagated up the tree. The dotted blue arrows indicate the update of the parent scores (e.g., the score updates from 77.47 to 154.31).
- After these iterations, the node with the highest cumulative value is locked into the chosen path, and then generates new child clips until reaching a total of **w_1=3** children, allowing the iterative generation process to continue.

SIGGRAPH ASIA 2025 HONG KONG 香港

AniMaker

Photograph Agent & Reviewer Agent



- The Rank 3 Candidate (Red Border): Excluded immediately because the blue plate comes from nowhere.
- The Rank 2 Candidate (Orange Border): Initially looks promising, but during the simulation, it reveals low continuous generation potential. Since the plate is too close to the bottom border of the screen, it, the next shot cannot fully show the scene of putting the plate on the table, causing the action to break down.

AniMaker Post-production Agent

Voiceover Scripting

Gemini 2.0 Flash generates a detailed script, specifying emotional tones and selecting voice profiles based on character attributes (e.g., age, gender).

Audio Synthesis

CosyVoice2 produces high-quality audio tracks, followed by automated verification to ensure content accuracy and duration consistency.

Film Assembly

The agent utilizes MoviePy to integrate video clips with the generated audio and subtitles, ensuring precise audio-visual synchronization.

SIGGRAPH ASIA 2025 HONG KONG 香港

AniMaker

Experimental Result

Table 3. VBench evaluation results, presenting scores for Image Quality (I.Q.), Semantic Consistency (S.C.), Background Consistency (B.C.), Animation Quality (A.Q.), Motion Smoothness (M.S.), Dynamic Degree (D.D.), and Average Rank (Rk. Avg. - the average ranking position across all models).

Model	I.Q.↑	S.C.↑	B.C.↑	A.Q.↑	M.S.↑	D.D.↑	Rk. Avg.↓
StableDiffusion+Cog.	75.52	78.05	85.27	59.61	97.63	33.58	5.83
StableDiffusion+Wan.	76.93	78.54	87.43	69.75	96.67	60.30	4.33
StoryAdapter+Cog.	76.17	72.17	88.03	63.55	98.16	26.71	5.33
StoryAdapter+Wan.	75.96	75.04	88.64	73.38	97.02	**84.73**	4.17
MovieAgent	72.09	68.61	79.84	55.40	99.01	35.44	6.33
MMStoryAgent	76.41	**87.27**	**90.74**	73.84	**99.80**	0.00	2.67
VoT	63.85	75.11	85.78	**74.91**	99.25	3.50	4.83
AniMaker(Ours)	**76.96**	84.27	89.06	69.79	98.50	66.97	**2.50**

Table 4. AniEval evaluation results, presenting scores for Overall Video Quality (O.V.Q.), Text-Video Alignment (T.V.A.), Video Consistency (V.C.), Motion Quality (M.Q.), and Total Performance.

Model	O.V.Q.↑	T.V.A.↑	V.C.↑	M.Q.↑	Total↑
StoryDiffusion+CogVideoX	46.54	86.05	47.14	70.35	56.75
StoryDiffusion+Wan 2.1	47.07	84.99	47.13	71.00	56.55
StoryAdapter+CogVideoX	56.76	**87.38**	55.89	69.73	63.95
StoryAdapter+Wan 2.1	60.39	86.99	51.41	72.11	62.37
MovieAgent	41.17	68.50	68.68	70.16	61.95
MMStoryAgent	47.93	75.27	63.54	61.39	62.79
VideoGen-of-Thought	66.17	72.95	65.42	66.72	66.93
AniMaker(Ours)	**81.87**	74.30	**79.35**	**72.66**	**76.72**

Table 5. Human rating results on a 1–5 scale, covering Character Consistency (C.C.), Narrative Coherence (N.C.), Physical-Law Adherence (P.L.), Script Faithfulness (S.F.), and Visual Appeal (V.A.).

Model	C.C.↑	N.C.↑	P.L.↑	S.F.↑	V.A.↑	Avg.↑
StoryDiffusion+CogVideoX	1.37	1.48	1.37	1.67	1.56	1.49
StoryDiffusion+Wan 2.1	2.00	1.82	1.82	2.00	2.11	1.95
StoryAdapter+CogVideoX	1.64	1.39	1.46	1.68	1.71	1.58
StoryAdapter+Wan 2.1	2.04	1.82	1.89	2.04	2.57	2.07
MovieAgent	1.19	1.26	1.44	1.26	1.48	1.33
MM_StoryAgent	1.62	1.62	1.72	1.83	2.24	1.81
VideoGenoT	1.67	1.74	1.78	1.74	2.26	1.84
AniMaker(Ours)	**3.44**	**3.24**	**3.04**	**3.08**	**3.28**	**3.22**

VBench Performance & Limitations: While AniMaker secures the best average rank (2.50) on VBench, the evaluation exposes VBench's critical flaw: it favors static consistency and fails to adequately assess dynamic storytelling animation.

AniEval Validation: Under the proposed AniEval framework, AniMaker achieves the top score of 76.72 (outperforming the runner-up by 14.6%) with exceptional Video Consistency. Crucially, AniEval aligns much closer to human evaluations than VBench.

Human Evaluation: Human raters consistently scored AniMaker highest across all metrics, particularly in Character Consistency and Narrative Coherence.

Lily bounces eagerly at the table, awaiting her mother's surprise.

Another Case Bird & Rabbit

Story Input

Once upon a time, there was a gifted **bird** named Blue. Blue could **whistle the best songs** in the forest. One day, Blue found a shiny black rock. It was **coal**. Blue took the coal to his friend, a **rabbit** named Bunny. Bunny liked the coal and used it to **draw pictures** on the ground. Blue and Bunny had a fun day playing with the coal and whistling songs.

Storyboard

Characters

Backgrounds

Keyframes

Scripts

- Character Description
- Background Description
- Visual Script
- Voiceover Script

Animation Movie

Blue Bird perches gently, **whistling a soft tune**, "What a beautiful forest day!"

Looking down, **she spots something below**. "This looks like perfect art material!"

Clutching the coal tightly in her beak, she soars toward the western horizon.

Landing near a burrow, she calls out, "Hello, Bunny! Surprise for you!"

Bunny Rabbit **emerges**, squinting at the bright sunlight and his unexpected visitor.

Bunny **picks up the coal**, ready to draw on the ground.

He begins **sketching swirling patterns on the ground**, coal in paw.

Together they make their way to **a flat rock beside the babbling stream** for another painting.

"Your pictures are amazing, Bunny!" **chirps** Blue Bird.

Finished drawing, Bunny **proudly holds up his artwork**. "Look what we created!"

Bunny **carefully tucks away the coal**, ready for more imaginative drawing next time.

Blue Bird soars skyward, leaving the glade bathed in golden light.

Blue Bird lands gracefully on a branch, surveying the clearing.

SIGGRAPH ASIA 2025 HONG KONG 香港

Takeaway Messages

• **MCTS-Gen**: MCTS-Gen strategy treats clip generation as a tree search, intelligently balancing exploration and exploitation.

• **AniEval**: Existing evaluation metrics like VBench are insufficient because they evaluate clips in isolation. To resolve this, we develope AniEval, the first framework specifically designed for multi-shot animation.

• **Future Work**: We will focus on updating the modular components with more advanced models to mitigate specific issues like model hallucination, aiming to bring AI-generated animation closer to professional standards.

If you have any question, please feel free to contact me

Personal Homepage: https://github.com/HaoyuanShi   E-mail: g1016015592@gmail.com

SIGGRAPH ASIA 2025 HONG KONG 香港

Thanks for your Listening!